

Starbucks Nutrition Data Analysis Report

1. Project Overview

Introduction:

With increasing consumer awareness regarding health and nutrition, understanding the nutritional composition of beverages has become important for both customers and businesses. This project analyses Starbucks beverage nutritional data to identify trends in calories, caffeine, sugar, protein, and beverage categories using Power BI.

Objective:

The primary objective of this project is to perform data analysis on Starbucks beverages and develop an interactive dashboard that enables users to:

- Compare nutritional values across beverage categories.
- Identify high-caffeine and high-calorie beverages.
- Analyse beverage distribution across categories.
- Support healthier beverage selection through data-driven insights.

Tools and Technologies Used:

- Microsoft Power BI
- Microsoft Excel / CSV Dataset
- Data Analysis Expressions
- Data Visualization Techniques

2. Dataset Description

The dataset contains nutritional information for Starbucks beverages, including multiple serving sizes and preparation styles.

Dataset Attributes:

Attribute	Description
Beverage Category	Beverage classification
Beverage	Beverage name
Beverage Prep	Preparation style/serving size
Calories	Total calories
Total Fat	Fat content
Sugars	Sugar content
Protein(g)	Protein content
Caffeine	Caffeine content
Sodium	Sodium content
Carbohydrates	Total carbohydrates

Dataset Summary:

- Total Beverage Records: 242
- Unique Beverages: 33
- Multiple Beverage Categories
- Multiple Beverage Preparation Types

3. Data Preparation

Data Cleaning:

The following preprocessing steps were performed:

- Verified data consistency.
- Checked for missing values.
- Validated numerical columns.
- Standardized beverage category names.
- Removed unnecessary fields for dashboard reporting.

Data Transformation:

DAX measures used in dashboard analytics

Total Beverages

Total Beverages = `DISTINCTCOUNT(starbucks[Beverage])`

Average Calories

Average Calories = `AVERAGE(starbucks[Calories])`

Average Sugar

Average Sugar = `AVERAGE(starbucks[Sugars (g)])`

Average Caffeine

Average Caffeine = `AVERAGE(starbucks[Caffeine (mg)])`

4. Dashboard Overview

The Power BI dashboard was designed to provide an interactive overview of Starbucks beverage nutrition. The dashboard displays key performance indicators:

Interactive filters for users to filter data:

- Protein Range
- Beverage Preparation Type

These filters dynamically update all visualisations.

KPI	Value
Total Beverages	33
Average Sugar	33.02 g
Average Calories	194.30
Average Caffeine	81 mg

5. Dashboard Analysis

Beverage Category Analysis:

The Average Calories by Beverage Category visualisation highlights calorie differences across beverage categories.

Findings:

- Smoothies contain some of the highest average calorie values.
- Frappuccino beverages are generally high in calories.
- Coffee-based beverages tend to have lower calorie content.

Insight: Customers seeking lower-calorie beverage options may prefer coffee-based drinks over blended beverages and smoothies.

Beverage Distribution Analysis:

The Beverage Category Distribution chart illustrates the proportion of beverages across categories.

Findings:

- Classic Espresso beverages represent a significant portion of the menu.
- Tazo Tea beverages contribute a moderate share.
- Smoothies represent the smallest category.

Insight: Starbucks' beverage portfolio is largely centred around coffee and espresso-based offerings.

Caffeine Analysis:

The Average Caffeine by Category chart compares caffeine levels across beverage categories.

Findings:

- Coffee beverages contain the highest caffeine levels.
- Espresso-based drinks also contribute significant caffeine content.
- Tea and smoothie categories generally contain lower caffeine values.

Insight: Customers looking for high-energy beverages are more likely to choose coffee and espresso-based products.

Top 5 Highest Caffeine Beverages: The dashboard identifies the beverages with the highest average caffeine content.

Beverage	Average Caffeine (mg)
Coffee	293.75
Caffè Americano	187.50
Frappuccino Light	173.33
Frappuccino Blended	133.75
Iced Beverages	123.13

Insight: Coffee significantly exceeds all other beverage categories in caffeine concentration, making it the strongest caffeine source in the dataset.

6. Key Insights

- High-Calorie Categories: Smoothies and Frappuccino beverages contain considerably higher calorie values compared to traditional coffee beverages.
- Caffeine Leaders: Coffee and Caffè Americano beverages provide the highest caffeine content among all beverage categories.
- Menu Composition: Espresso and coffee-based beverages dominate Starbucks' product offerings.
- Nutritional Variation: Nutritional values vary significantly across beverage preparations and serving sizes.
- Consumer Choice: Customers can use nutritional information to select beverages aligned with their dietary goals and caffeine preferences.

7. Business Recommendations

1. For Health-Conscious Customers
 - ➔ Promote lower-calorie beverage options.
 - ➔ Highlight beverages with reduced sugar content.
 - ➔ Encourage awareness through nutritional labelling.
2. For Energy-Focused Customers
 - ➔ Recommend coffee and espresso beverages due to higher caffeine levels.
 - ➔ Provide caffeine information prominently within ordering platforms.

3. For Product Development

- Expand low-sugar beverage alternatives.
- Introduce more high-protein beverage options.
- Develop healthier variants of popular blended drinks.

8. Conclusion

This project successfully analysed Starbucks beverage nutritional data using Power BI and DAX measures. Through interactive visualisations and KPI tracking, valuable insights were obtained regarding calorie content, caffeine concentration, sugar levels, and beverage distribution.

The dashboard provides an intuitive platform for exploring beverage nutrition and supports informed decision-making for both customers and business stakeholders. The analysis demonstrates how data visualisation can transform raw nutritional information into meaningful business insights.